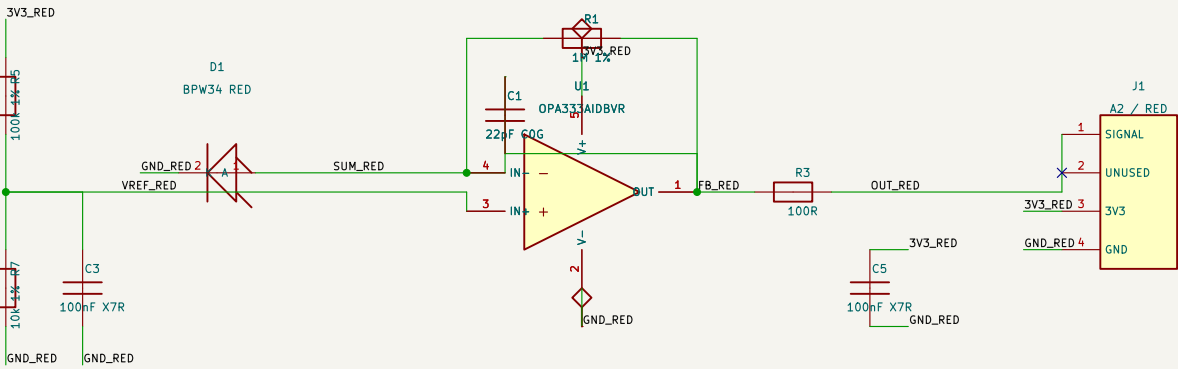
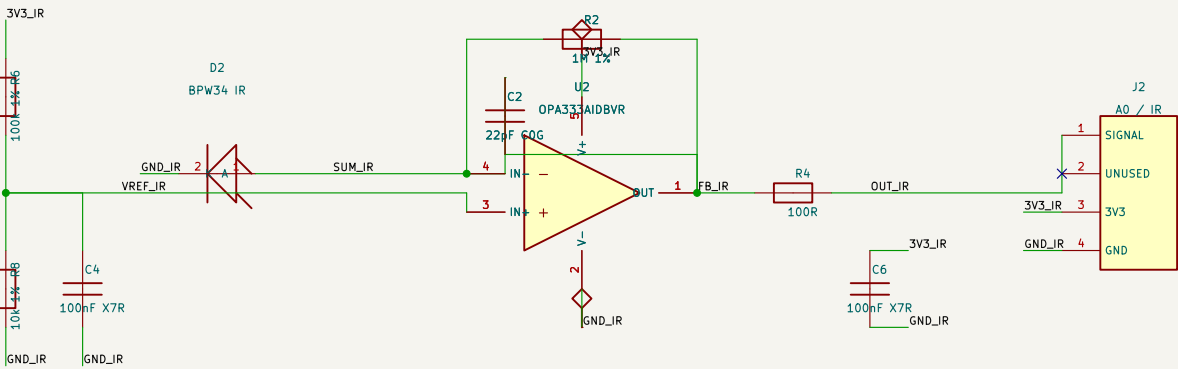


RED CHANNEL – BPW34 –> 1 Mohm TIA –> A2 / RED

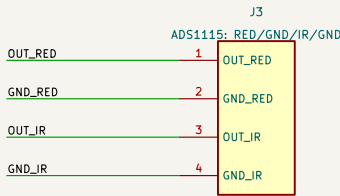


Grove: 1=signal, 2=NC, 3=3V3, 4=GND

IR CHANNEL – BPW34 –> 1 Mohm TIA –> A0 / IR



Grove: 1=signal, 2=NC, 3=3V3, 4=GND



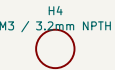
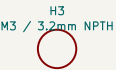
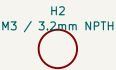
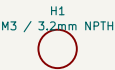
J3 -> ADS1115: each signal travels with its own channel ground

Power the ADS1115 separately. J3 does not provide VCC.

Default gain: 1 MOhm. Approx. $V_{OUT} = 0,30 \text{ V} + I_{PD} \times 1 \text{ MOhm}$.

C1/C2 = 22 pF COG stability capacitors; all 100 nF capacitors are non-polar ceramic.

OPA333 SOT-23-5 may be replaced by pin-compatible MCP6001T-I/OT for lower cost.



70 x 32 mm / Grove A2 RED + A0 IR / ADS1115 output
Nguyen Nhat Huy – Pham Thanh Vy

Sheet: /
File: bpw34_receiver.kicad_sch

Title: DATN: PPG-Simulator – Dual BPW34 receiver

Size: A3 Date: 2026-09-24

Rev: 1.1

KiCad E.D.A. 10.0.6

Id: 1/1